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Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Space Human Factors Engineering		
Joint Agency Name:		**TechPort:**	No
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Task Description:	Although returning crewmembers have higher incidence of disc herniation post-flight, it is unknown if this increased incidence is related or contributes to an increased risk of injury due to dynamic loads. The aims are as follows: 1. Mine existing human spaceflight data on the incidence and severity of vertebral disc herniation after spaceflight. 2. Analyze data (perform meta-analysis, statistics, etc.). 3. Determine the correlation of herniation risk to flight duration and landing impact severity.		

Rationale for HRP Directed Research:	Insufficient time for solicitation: The results of this study will inform Deconditioning Factors to be applied to Injury Assessment Reference Values (IARVs) to then be used by the Multi-Purpose Crew Vehicle (MPCV) Program for incorporation into future design and analysis cycles. This information must be delivered to MPCV by mid FY16, which requires that this task commence no later than the beginning of FY14 to address any follow on actions as a result of the finding. Therefore, there is inadequate time to solicit this research.
Research Impact/Earth Benefits:	Understanding of the effects of spaceflight on disc herniation may shed light on injury mechanisms on Earth, thereby informing therapies.

A previous study reported that the risk of developing a herniated nucleus pulposus (HNP) was higher in U.S. astronauts who had flown at least one mission, as compared with those in the NASA Astronaut Corps who had not yet flown. However, the study did not account for the cumulative effect of multiple missions. While many HNPs occurred well into astronauts' careers or in some cases years after retirement, the higher incidence of HNPs relatively soon after completion of space missions suggests that spaceflight does increase the risk of HNP. To formally investigate this possibility, we used a competing-risks survival model to analyze reports and dates of first HNP occurrences from 330 U.S. astronauts taking part in 745 space missions over 55 years. In this model, the outcome of interest, time from selection into the Astronaut Corps to first report of HNP is modeled as the minimum of several event times, one following each space mission plus one influenced by astronaut training and general lifestyle irrespective of participation in actual missions. We found that spaceflight experience significantly increased the risk of HNP; however, long-duration missions and capsule landings were not found to have a significant effect.

Background

Following extensive reports of back pain occurring in astronauts during or after many NASA spaceflight missions, Johnston, et al. [1] launched a retrospective study to compare incidence of herniated nucleus pulposus (HNP) in astronauts with that of a matched control population, and also studied whether the time to occurrence of HNP (starting with selection into the Astronaut Corps) differed between astronauts who had not yet participated in a mission and those who had flown at least one mission. Based on analysis of data from 321 U.S. astronauts over the years 1959–2006, astronauts had a higher incidence rate of HNP than matched controls and, for astronauts specifically, the hazard function associated with time to HNP was increased after a first spaceflight mission. However, in this first study the time to HNP was modeled the same way no matter how many years had elapsed between the date of a first mission and the occurrence of HNP, which in some cases was many missions and/or years later. In a related study, Kerstman, Scheuring, et al. [2] analyzed reports of in-flight back pain (not necessarily caused by HNP) and reviewed the efficacy of various treatments. Recently (2014), because of concern about the possible mission-threatening potential of both in-flight back pain and actual HNP, the NASA Human System Risk Board (HSRB) requested an updated study. The HSRB is tasked with identifying hazards to human spaceflight, caused by altered gravity, exposure to radiation, etc. [3]. In particular, the HSRB wanted to know if reports of HNP after missions could be attributed to the missions themselves or was it mainly because members of the Astronaut Corps (as are many non-astronauts) are continually at risk of developing a HNP as they age, regardless of when and how many missions they flew. In response to the HSRB directive, this study was launched in mid-2014, analyzing reports of HNP updated to include 330 U.S. astronauts through February 2014. Here, the main objective was to tease out the direct effect of recent space missions from the chronic effect of being in the Astronaut Corps, either active or retired. In the process, we investigated whether multiple missions, sex, age, vehicle landing dynamics, and flight duration affect the risk of developing a HNP.

Relevance

The Occupant Protection Discipline is interested in determining if deconditioning, as a result of spaceflight, leads to increased risk of intervertebral disc damage/herniation. A study by Johnston, et al. [1] compiled disc herniation incidence data from 321 astronauts from April 1959 to December 2006 and concluded that astronauts have a 4.3 times greater risk of disc herniation than control subjects. Although these results indicate that astronauts have increased risk of herniation, among other things, the paper did not examine the contribution of mission duration or landing impact on likelihood of disc damage. It is the goal of this Disc Herniation Risk Analysis study to compare disc herniation rates between capsule and Shuttle landings, as well as short- and long-duration missions. If the incidence of disc herniation is significantly higher for crewmembers who landed in capsule spacecraft and/or who were on long-duration missions, then future work to investigate the intervertebral disc would be warranted. If no significant difference is found, then deconditioning of the intervertebral disc most likely does not play a significant role in impact injury risk.

The study will contribute toward closing the following Integrated Research Plan (IRP) gap: OP-03: We need to understand the effect of spaceflight deconditioning on injury risk due to dynamic events during the mission (< 0.5 s in duration).

Methods

The study was done using data provided by the NASA Lifetime Surveillance of Astronaut Health covering the period from 1959 to February 2014. The data consisted of (a) dates that U.S. astronauts were selected into the Astronaut Corps, (b) when they began and finished spaceflight missions, (c) basic mission characteristics such as length of mission and type of landing vehicle (Shuttle or capsule), (d) dates at which HNP diagnoses were made, and (e) basic demographic data (e.g., gender, age, weight, height). HNP diagnoses were confirmed by MRI or CT scans, myelography, operative findings, or through clinical corroboration by a neurologist or neurosurgeon. Pre-existing cases of HNP diagnosed at or before the time an astronaut was selected into the Astronaut Corps were ignored. Overall, we observed 51 cases of first HNP in 330 U.S. astronauts studied over time spans ranging from 0.2 to 52.0 years. In many cases, astronauts returned for yearly physical examinations even after retirement; others were lost to follow-up after retirement or were deceased. The number of missions studied per astronaut ranged from 0 (those selected into the Astronaut Corps but who hadn't yet participated in a mission) to 7 comprising a total of 745 astronaut missions. After including one observation of HNP status for each astronaut just prior to his/her first mission, the resulting total number of observations for potential analysis becomes 1075. An additional 44 observations (4 HNPs) were not analyzed because this data was collected after first HNPs had already occurred. In addition to HNP status, we also collected corresponding observations of several explanatory variables for subsequent modeling.

Statistical Methods

Task Progress:

The data show declining HNP incidence rates as time elapses after a mission, which might suggest that spaceflight is associated with HNP risk. Indeed, it is this trend that has given rise to NASA directives to study the issue. However, it does not take into account the lengths of time that astronauts were studied, data censoring, or that the hazard (instantaneous risk given no previous HNP) may not be constant. It also ignores possible effects of multiple missions. To more properly assess whether spaceflight plays a role in development of HNPs, we used a competing-risks survival model to estimate the distribution of Tmin, the time from selection into the Astronaut Corps until a first diagnosis of HNP by direct modeling of its hazard function in terms of several explanatory variables. The explanatory variables fall into two categories: flight-related and demographic. The flight-related explanatory variables are number of missions, timing of missions, mission duration(s), type of landing vehicle(s), and high-performance, jet-aircraft-pilot experience. For purposes of this study, “long-duration” missions were those flown on Skylab, MIR, or ISS (International Space Station). Others were considered “short-duration” missions. Landing vehicles were classified into “STS” (Shuttle) or “capsule” (all others). The demographic explanatory variables are age, sex, weight, height, and BMI (body mass index).

Under the competing-risks model, Tmin is modeled as the minimum occurrence time of an HNP given the risk of occurrence attributable to the astronaut training programs and typical astronaut lifestyle, and the effects of spaceflight. Except for one situation, we can only observe T_min, and not know the underlying cause. Thus, for example, if a HNP diagnosis was made after an astronaut’s third mission, we cannot tell if the contributing factor occurred during that mission, during a previous mission, or was not even mission-related. Instead, we can only assess cause probabilistically.

Results

Occurrence

Of the 51 observed HNP, 19 were cervical, 3 were thoracic, and 29 were lumbar. In addition, 46 of the 51 were observed after long-duration missions, and 40 of the 51 were observed after Space Shuttle missions.

Other than a small effect of age at selection, there was little evidence that mission duration, type of landing vehicle, or other demographic variables contributed to the model fit. In general the HNP risk after missions drops off sharply. However, the actual form of the distributional functions depends on the number and timing of missions, even in the no-covariate model. In a hypothetical crewmember with multiple flights, despite a sharp drop in the individual hazards related to each spaceflight, there would be enough carryover from one flight to the next in this hypothetical case to produce a noticeable upward trend in HNP risk as the number of flights increases.

Attribution of HNPs to Spaceflight

One of the most important questions addressed by this study was to separate the effects of spaceflight from those of the general astronaut training and lifestyle. Using point estimates of the parameters to the 40 cases of HNP observed after at least one space mission, yielded values of P(S|?T_min=t?^*) ranging from 0.037 to 0.970 depending on when the HNP was reported. As expected, this probability is highest for HNP’s reported shortly after the latest mission for each astronaut.

A total of 745 out of 789 astronaut-missions took place before crewmembers’ first diagnosed HNP. Of these, 58 (7.8%) were “long duration” and 98 (13.2%) were capsule landings. No evidence of an effect of either flight duration or landing vehicle on HNP propensity after spaceflight was seen. There was no strong evidence that sex, height, weight, BMI, or a history of high-performance jet aircraft piloting had an effect on HNP risk either. However, astronauts that were older at the time of selection appeared to have higher risk (P = 0.038). In addition, pre-Shuttle astronauts had generally lower risk of HNP (P = 0.012).

Discussion

Analysis of the data revealed clear evidence that spaceflight is associated with increased risk of HNP, thus supporting the conjecture suggested by a higher incidence of HNPs shortly after missions. In arriving at this finding we fit a survival model that took into account differences in type, number, and timing of missions as well as the periods of observation for each astronaut. The model-based conditional probabilities that each of the 44 HNPs that occurred after at least one mission were attributable to spaceflight, ranged from 0.97 (shortly after a mission) to about 0.2 or lower (at least 30 years after selection). The average value of these probabilities was 0.44.

Other than a detrimental effect of initial age (i.e., at selection), we did not find evidence that HNP risk was affected by demographic factors such as gender, height, weight, or whether an astronaut had experience piloting a high-performance jet aircraft. It did appear that astronauts from the pre-Shuttle era were at lower risk of eventually developing an HNP (P = 0.012). Finally, we did not find evidence that either mission duration or type of landing vehicle had an effect.

Limitations

Because this was an observational study it is difficult to separate out the effects of the many spaceflight and demographic factors on HNP risk or to claim causality. In particular, we had no control over when long-duration or capsule-landing missions occurred, thus creating substantial confounding of these factors with HNP reporting and diagnosis practices as well as changing criteria for astronaut selection since the pre-Shuttle era. Also, HNP time of incidence was recorded at the time of diagnosis, not at the time of occurrence.

Because the data span the entire Astronaut Corps, effects from improved spaceflight deconditioning countermeasures may obscure the risk of developing an HNP, particularly related to mission length. A majority of the long-duration missions occurred in the past 25 years, when countermeasures have been implemented. In addition, the relatively low numbers of long-duration flights and female astronauts adversely affects the power of tests for these effects.

Finally, the current study only examines data from U.S. astronauts. Supplementing the data with HNP reports from other International Partners could allow more insight into these effects.

Future Work

To better assess the effects of spaceflight on HNP risk, additional crewmember data would be ideal. In addition to the U.S. astronauts included in this study, additional information may be available from the international partners, which could increase the dataset substantially. Additional countermeasures for crewmembers immediately after landing may also be advised to prevent HNP occurrence at the times of highest risk. Finally, continued surveillance of crewmembers

	after spaceflight could allow a better understanding of this trend. These results may also be beneficial to current studies of the intervertebral disc and additional analysis of these data in concert with the data from the current studies may improve our understanding of the mechanism of HNP after spaceflight. Acknowledgements The author would like to thank Dr. Alan H. Feiveson from the NASA Biostatistics Laboratory for his statistical modeling and analysis of the data. This work is supported by the Human Research Program (HRP) under contract NAS9-02078. References [1] Johnston S, Campbell M, Scheuring R, Feiveson A. Risk of Herniated Nucleus Pulposus Among U.S. Astronauts. Aviat Space Environ Med 2010;81:566-74. [2] Kerstman EL, Scheuring RA, Barnes MG, DeKorse TB, Saile LG. Space adaptation back pain: a retrospective study. Aviat Space Environ Med 2012;83(1):2-7. [3] National Aeronautics and Space Administraion. Human Research Program Architecture. [cited 2015; Available from: http://humanresearchroadmap.nasa.gov/
Bibliography Type:	Description: (Last Updated: 12/29/2020)
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Abstracts for Journals and Proceedings	Feiveson A, Mendez C. Somers J. "Assessing the Effect of Spaceflight on the Propensity for Astronauts to Develop Disc Herniation." Presented at the 2015 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 13-15, 2015. 2015 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 13-15, 2015. , Jan-2015